

FoodSense: A Multimodal IoT–AI System for Early Food Spoilage Prediction Using Gas Sensing, Visible Spectroscopy, and Vision

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Abstract—Food spoilage accounts for nearly 19% of global food production loss, creating pressing food-security and economic consequences. Existing automated detection systems either rely on expensive multi-sensor electronic noses or yield only binary “fresh/spoiled” labels without temporal context. This paper presents FoodSense, a low-cost, multimodal IoT–AI pipeline that integrates (i) an MQ135 metal-oxide gas sensor with on-edge temperature-humidity drift compensation, (ii) a DIY RGB-LED/LDR visible spectrophotometer for pigment-level analysis, and (iii) a server-side image-analysis branch using a fine-tuned CNN and Vision Transformer. A Multi-Layer Perceptron (MLP) trained on the UCI Gas Sensor Array Drift Dataset achieves 96.84% accuracy and an F1-score of 87.28% for VOC-elevation detection. An attention-augmented Long Short-Term Memory (LSTM) network provides continuous shelf-life regression with a Mean Absolute Error of ± 4.0 hours on a proof-of-concept synthetic time-series. A virtual spectrometer calibration model maps RGB-LDR signals to visible wavelengths with $R^2 = 0.9625$, and a pigment-degradation classifier distinguishes Fresh, Ripe, and Spoiled stages at 95.60% accuracy. The full system is deployed as an open-source FastAPI web application on Hugging Face Spaces, exposing real-time WebSocket telemetry and a mobile-responsive dashboard.

Index Terms—food spoilage detection, IoT sensor fusion, MLP gas classifier, attention LSTM, visible spectroscopy, multimodal AI, edge computing, shelf-life prediction

I. INTRODUCTION

The United Nations Environment Programme estimates that approximately 19% of all food produced globally is wasted before consumption [1]. Spoilage-driven waste has downstream consequences for greenhouse gas emissions, economic loss, and food insecurity. The commercial solution—an electronic nose (e-nose) comprising a calibrated array of metal-oxide sensors—costs upward of several thousand USD and is confined to laboratory settings [2]. Consumer-grade systems are

therefore limited to manual inspection, a subjective and error-prone process.

Recent advances in edge computing and deep learning have opened the possibility of deploying AI-driven spoilage detectors on low-cost hardware. However, the majority of prior work falls into one of two categories: (a) single-modality systems that either analyse gas-phase volatile organic compounds (VOCs) or classify visual ripeness, but not both; and (b) static classifiers that report an instantaneous label without any estimate of remaining shelf life.

This paper addresses both gaps through the following contributions:

- **On-edge drift compensation.** A closed-form temperature-humidity correction formula is executed directly on the Arduino microcontroller, eliminating the need for post-hoc recalibration.
- **Continuous shelf-life regression.** A 2-layer, attention-augmented LSTM processes a rolling 30-reading window to predict remaining shelf life in hours, rather than outputting a binary label.
- **Multimodal architecture.** Three parallel inference branches—gas/climate, spectroscopy, and vision—contribute to a unified freshness score served over WebSockets.
- **Virtual spectrophotometry.** A machine-learning calibration model maps cheap RGB-LED/LDR signals to continuous wavelength (nm) and absorbance values, realising a sub-\$5 spectrometer.

II. RELATED WORK

Gas-sensor spoilage detection. Vergara et al. [2] collected a 36-month benchmark demonstrating significant drift in metal-oxide sensors. Huerta et al. [3] quantified the contribution of

temperature and humidity to drift in open-sampling e-noses. Yi et al. [4] proposed discriminant subspace learning for batch drift compensation, while Chang et al. [5] analysed temporal and spatial artifacts in public gas datasets. Guo et al. [6] classified *Penicillium expansum* spoilage in apples using an e-nose combined with chemometrics.

Spectroscopy and colorimetry. Fodor et al. [7] demonstrated that NIR absorption peaks at specific wavelengths directly indicate chlorophyll degradation and melanin formation—markers of spoilage. Albert et al. [8] showed that Arduino-controlled RGB LEDs paired with LDR photodetectors can perform quantitative spectrophotometry. Ren et al. [9] extended this approach with a 3D-printed modular spectrometer for education and research.

Vision and deep learning. Amin et al. [10] demonstrated CNN-based automatic fruit freshness classification with high accuracy. Gao et al. [11] applied Vision Transformers (ViT) for food image classification, achieving state-of-the-art results. Nguyen et al. [12] designed an optical fruit-ripeness detector for automated sorting.

Temporal modeling. Wu et al. [13] proposed the GCNN-LSTM-AT network for tangerine quality prediction using Vis-NIR spectroscopy. Saha et al. [14] built an IoT-based fruit lifespan detection system at the network edge.

FoodSense advances the state-of-the-art by combining all three modalities—gas, spectroscopy, and vision—in a single low-cost platform with on-edge drift correction and continuous shelf-life output.

III. SYSTEM ARCHITECTURE

FoodSense is structured as a three-tier system: (1) an *edge hardware node* for data acquisition, (2) a *FastAPI backend* for AI inference, and (3) a *web dashboard* for visualisation and interaction.

A. Edge Hardware Node

The hardware node houses two sensing sub-systems in a compact enclosure:

Gas/Climate Sensor Node. An MQ135 metal-oxide gas sensor [3] reads ambient VOC concentration via the Arduino ADC. A DHT11 sensor supplies temperature (T , °C) and relative humidity (H , %). On-device drift compensation is applied before serial transmission:

$$\text{corr} = 1 + 0.02(T - 20) + 0.01(H - 65) \quad (1)$$

$$R_{s,\text{corr}} = \frac{R_s}{\text{corr}} \quad (2)$$

The corrected resistance $R_{s,\text{corr}}$ is then converted to an estimated ammonia-sensitive VOC concentration (ppm) using the standard power-law sensor model and transmitted via USB serial as comma-separated values (CSV) at 2-minute intervals.

DIY Visible Spectrophotometer. A common-cathode RGB LED illuminates a cuvette containing the test solution; an LDR photodetector measures transmitted intensity at red (660 nm), green (540 nm), and blue (430 nm) channels. The assembly is housed in a 3D-printed dark chamber to eliminate ambient

light interference. Raw ADC readings are transmitted alongside the gas data to the host computer.

B. Backend AI Engine

The FastAPI server exposes three REST/WebSocket API routes, each connected to dedicated AI models:

Gas API Route. Feeds temperature-compensated VOC readings to (a) an MLP binary gas classifier and (b) an attention-augmented LSTM for shelf-life regression.

Spectroscopy API Route. Accepts RGB-LDR absorbance values and routes them to (a) a Virtual Spectrometer Calibration MLP and (b) a Pigment Degradation Classifier MLP.

Vision API Route. Accepts JPEG/PNG frames and runs inference through a fine-tuned EfficientNet-B0 CNN for spoilage-state classification and a Vision Transformer (ViT-B/16) for quality regression, both pre-trained on ImageNet and fine-tuned on a combined Kaggle fruit-freshness dataset [10].

C. Web Dashboard

A Tailwind CSS + JavaScript single-page application is hosted on Hugging Face Spaces. The dashboard maintains a persistent WebSocket connection for live telemetry, renders chart.js time-series plots, and supports image upload for on-demand visual analysis.

IV. AI MODEL DESIGN

A. MLP Gas Classifier

The MLP is trained on the UCI Gas Sensor Array Drift Dataset [2], which contains 13,910 real sensor readings from a 36-month experiment. The binary label indicates elevated VOC exposure (ammonia, acetone, ethylene) versus baseline air.

Architecture.

Linear(128, 64) → BN → ReLU → Dropout(0.3)
→ Linear(64, 32) → BN → ReLU → Dropout(0.2)
→ Linear(32, 16) → ReLU
→ Linear(16, 1) → BCEWithLogitsLoss

Training. Adam optimiser, learning rate 10^{-3} , weight decay 10^{-4} , 80/20 stratified split, `pos_weight=7.5` for class imbalance correction.

B. Attention-Augmented LSTM

Spoilage is a dynamic process; the LSTM captures the temporal progression of VOC signals across a 1-hour rolling context window (30 readings, one per 2 min).

Architecture.

LSTM(hidden = 64, layers = 2, dropout = 0.2)
→ Temporal Attention Layer
→ Linear(64, 32) → ReLU
→ Linear(32, 1) → Sigmoid × 48 h

Training. Huber loss, ReduceLROnPlateau scheduler, gradient clipping (max_norm=1.0), target normalised to [0, 1].

C. Virtual Spectrometer Calibration MLP

The calibration model maps $(R_{\text{raw}}, G_{\text{raw}}, B_{\text{raw}})$ triplets from the LDR to continuous wavelength (nm) and absorbance (Abs) values. It is trained on an experimental potassium permanganate (KMnO_4) scanning dataset generated using the DIY spectrophotometer.

D. Pigment Degradation Classifier

A 3-class MLP trained on laboratory spectral measurements classifies produce into *Fresh*, *Ripe*, and *Spoiled* categories by evaluating optical absorption at the chlorophyll bands (430 nm, 660 nm) and the enzymatic-browning band (540 nm).

V. EXPERIMENTAL RESULTS

A. MLP Gas Classifier

Table I reports performance on the 20% hold-out test split (2,782 samples). The model achieves 96.84% accuracy with strong recall, ensuring that genuine VOC elevations (indicative of active spoilage) are rarely missed.

TABLE I
MLP GAS CLASSIFIER — TEST SET PERFORMANCE

Metric	Score
Accuracy	96.84%
Precision	82.97%
Recall	92.07%
F1-Score	87.28%
True Negatives	2392
False Positives	62
False Negatives	26
True Positives	302

The 26 false negatives represent 8.0% of positive instances, a clinically meaningful but operationally acceptable miss rate for an early-warning system where false positives merely trigger early inspection.

B. LSTM Shelf-Life Predictor

The LSTM is evaluated on a synthetic 48-hour spoilage time-series (1,440 samples at 2-min intervals) using a chronological 90/10 train-test split. This constitutes a proof-of-concept evaluation; real-world performance will require walk-forward cross-validation on collected hardware data.

TABLE II
LSTM SHELF-LIFE PREDICTOR — PROOF-OF-CONCEPT METRICS

Metric	Score
MAE	± 4.00 hours
RMSE	4.11 hours

The temporal attention mechanism allows the model to assign higher weight to recent VOC spikes that signal accelerating spoilage, improving short-horizon predictions.

C. Virtual Spectrometer Calibration

An R^2 of 0.9625 for wavelength prediction confirms that the discrete RGB emission profiles are sufficient to reconstruct the continuous visible spectrum with high fidelity.

TABLE III
VIRTUAL SPECTROMETER CALIBRATION MODEL METRICS

Target	MSE	R^2
Wavelength (nm)	182.29	0.9625
Absorbance	0.0088	0.7240

D. Pigment Degradation Classifier

TABLE IV
PIGMENT CLASSIFIER — PER-CLASS REPORT

Class	Precision	Recall	F1
Fresh	0.98	0.97	0.97
Ripe	0.94	0.95	0.94
Spoiled	0.95	0.95	0.95
Accuracy	95.60%		

Monitoring absorption at 430 nm and 660 nm (chlorophyll *a* and *b*) and at 540 nm (oxidative browning products) provides a robust three-point chemical fingerprint sufficient to distinguish all three ripeness stages.

VI. DEPLOYMENT AND DASHBOARD

The full FoodSense stack is containerised and deployed on Hugging Face Spaces via a GitHub Actions CI/CD pipeline. Model updates are automatically pushed to the live endpoint. The dashboard provides:

- **Live sensor tab:** scrolling VOC, temperature, and humidity charts with shelf-life countdown updated every 3 s.
- **Evaluation tab:** confusion matrix, precision–recall curves, and calibration plots.
- **Image analysis tab:** drag-and-drop JPEG upload with CNN and ViT inference results displayed within ≈ 1 s.
- **Architecture tab:** interactive multiframe diagram illustrating the six-model AI engine.

VII. CONCLUSION

FoodSense demonstrates that a sub-\$30 hardware platform—an Arduino, an MQ135 gas sensor, a DHT11 humidity probe, and a DIY RGB-LED/LDR spectrophotometer—can support a production-grade multimodal AI pipeline for early food spoilage detection. On-edge temperature-humidity compensation eliminates the single largest source of metal-oxide sensor drift without any server-side post-processing. The parallel MLP and attention-LSTM inference tracks provide both instantaneous VOC classification and continuous shelf-life regression, addressing a critical gap in existing consumer-grade systems.

Future work will: (i) collect real sensor time-series to validate and retrain the LSTM with walk-forward cross-validation; (ii) implement model compression (quantisation, pruning) to deploy the CNN and ViT branches directly on a Raspberry Pi; and (iii) expand the food-type coverage to include grains, dairy, and processed foods.

REFERENCES

- [1] M. A. Davidescu, C. Pânzaru, B. M. Mădescu, I. Poroşnicu, C. Simeanu, A. Usturoi, M. Matei, and M. G. Doliş, "Advances and challenges in smart packaging technologies for the food industry: Trends, applications, and sustainability considerations," *Foods*, vol. 14, no. 24, p. 4347, 2025.
- [2] A. Vergara, J. Fonollosa, L. Fernandez, S. Gutierrez, and S. Marco, "Gas sensor array drift dataset," 2012. [Online]. Available: <https://archive.ics.uci.edu/dataset/224/gas+sensor+array+drift+dataset>
- [3] R. Huerta, T. S. Mosqueiro, J. Fonollosa, N. F. Rulkov, and I. Rodriguez-Lujan, "Online decorrelation of humidity and temperature in chemical sensors for continuous monitoring," *Chemometrics and Intelligent Laboratory Systems*, vol. 157, pp. 169–176, 2016.
- [4] Z. Yi, W. Shang, T. Xu, S. Guo, and X. Wu, "Local discriminant subspace learning for gas sensor drift problem," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 52, no. 1, pp. 247–259, 2022.
- [5] I.-S. Chang, S.-W. Byun, T.-B. Lim, and G.-M. Park, "A study of drift effect in a popular metal oxide sensor and gas recognition using public gas datasets," *IEEE Access*, vol. 11, pp. 26 383–26 392, 2023.
- [6] Z. M. Guo *et al.*, "Classification for *Penicillium expansum* spoilage and defect in apples by electronic nose combined with chemometrics," *Sensors*, vol. 20, no. 7, p. 2130, 2020.
- [7] M. Fodor, A. Matkovits, E. L. Benes, and Z. Jókai, "The role of near-infrared spectroscopy in food quality assurance: A review of the past two decades," *Foods*, vol. 13, no. 21, p. 3501, 2024.
- [8] D. R. Albert, M. A. Todt, and H. F. Davis, "A low-cost quantitative absorption spectrophotometer," *Journal of Chemical Education*, vol. 89, no. 11, pp. 1432–1435, 2012.
- [9] Z. Ren *et al.*, "Multirole 3d-printed modular spectrometer for various teaching spectral experiments in the classroom and at home," *Journal of Chemical Education*, vol. 100, no. 9, pp. 3622–3631, 2023.
- [10] U. Amin, M. Shahzad, A. Shahzad, M. Shahzad, U. Khan, and Z. Mahmood, "Automatic fruits freshness classification using cnn and transfer learning," *Applied Sciences*, vol. 13, no. 14, p. 8087, 2023.
- [11] X. Gao, Z. Xiao, and Z. Deng, "High accuracy food image classification via vision transformer with data augmentation and feature augmentation," *Journal of Food Engineering*, vol. 360, p. 111833, 2024.
- [12] D.-L. Nguyen, X.-T. Vo, A. Priadana, M. D. Putro, and K.-H. Jo, "Fruit ripeness detector for automatic fruit classification systems," *Proceedings of the IEEE International Symposium on Industrial Electronics (ISIE)*, pp. 1–6, 2024.
- [13] Y. Wu, X. Zhu, Q. Huang, Y. Zhang, J. Evans, and S. He, "Predicting the quality of tangerines using the gcnn-lstm-at network based on vis-nir spectroscopy," *Applied Sciences*, vol. 13, no. 14, p. 8221, 2023.
- [14] A. Saha, L. Ali, R. Rahman, T. Hossain, S. J. M. B. Shamim, M. A. A. Supto, M. Shidujaman, M. R. Uddin, and M. Hasan, "Iot based fruit quality inspection and lifespan detection system," *Proceedings of the 2023 7th International Conference on Computing, Communication, Control and Automation (ICCUBEA)*, pp. 1–6, 2023.